RAW WATER DISTRIBUTION
2000-F24-PID-PR-10002
RAW WATER
72-2513-BW-PE8-110
WI13121B
2000-F16-PID-PR-10001
SIGNAL
WI18027
2000-F18-PID-PR-10001
SIGNAL
FI17003
2000-F17-PID-PR-10001
SIGNAL
SCHEME WATER DISTRIBUTION
2000-F24-PID-PR-10006
SCHEME WATER
72-1060-PSW-SS2-100
PROCESS WATER/REAGENTS/
TRASH SCREENS AIR RECEIVER
2000-F24-PID-PR-10013
PLANT AIR
72-1061-IA-MSG3-25
72-0992-IA-MSG3-25
NOTE 2
HS 17003A
FIC 17003
f(x)
100VBFD1L2
F72-13112
NOTE 8
100VBFE1S2
F72-13125
SP MS-001
SET @ 600kPa
25VB03S1 F72-12158
25VB03S1 F72-12159
PDI 72064
XV 72008
DETAIL 8
HS 72003
XV 72003
F72-XM-441
FLOCCULANT SILO DEHUMIDIFIER
DN100 MALE CAMLOCK C/W CAP
OD12 PUSH FITTING
DN100
FLOCCULANT DELIVERY
NOTE 4
SP HS-076
WI 72001
PRV 72002
FC
SV 72004C
SV 72004B
SV 72004A
F72-DC-403
FLOCCULANT SILO DUST COLLECTOR
XA 72061
F72-DC-403 SYSTEM FAULT
XC 72063
F72-DC-403 SYSTEM ENABLE
LI 72001
LI 72001
HH H
LL
LT 72001
CCTV
F72-CAM-001
DETAIL 8
HS 72008
NOTE 6
SIREN
YA 72001
YA 72001
F72-SI-401
FLOCCULANT SILO
FE 72006
FIT 72006
FIC 72006
FCV 72007
PT 72014
PI 72014
FE 72005
FIT 72005
FIC 72005
F72-AT-401
FLOCCULANT SILO ACTIVATOR
FCV 72013
PI 72015
UNLOAD START HS 72021
UNLOAD STOP HS 72022
UNLOAD READY XI 72023
NOTE 5
PAH 72016
TAH 72017
PSH 72016
TSH 72017
F72-BW-411
FLOCCULANT SILO BLOWER
F72-FE-411
FLOCCULANT FEEDER
F72-HT-441
FLOCCULANT HEATED CONE
F72-VT-100
FLOCCULANT EDUCTOR & VENTURI MODULE
F72-XM-410
FLOCCULANT WETTING HEAD
FSL 72018
FSL 72019
LT 72031
LI 72031
HH
FAL 72018
FAL 72019
O/F
DN100
F72-TK-465
FLOCCULANT MIXING TANK
F72-AG-465
FLOCCULANT MIXING TANK AGITATOR
F72-PK-401
FLOCCULANT MIXING PACKAGE
VENDOR PACKAGE
2000-ME029-E02-10001
2000-ME029-E02-10002
SET AT 600 kPag
PRV 72020
100x80
TO GRADE
F72-PP-690
FLOCCULANT TRANSFER PUMP
DN200
LI 72032
LT 72032
72-0971-FL-MS1-200
72-0972-FL-MS1-100
O/F
100VBD1C1 F72-12160
200VBD1S3 F72-12161
F72-TK-466
FLOCCULANT STORAGE TANK
100VBD1C1 F72-12162
200-MECH-MS1
72-2202-FL-MS1-200
50VB03S1 F72-12163
LI 72071
LIT 72071
LE 72071
80VBFD1L2 F72-12443
80x65
NOTE 1
F72-PS-418
FLOCCULANT AREA SUMP PUMP
MS1
PE8
72-1073-FL-MS1-80
25VB03S1 F72-12444
SP MC-001

NOTES:

1. FLANGED ASME B16.5 CLASS 150.
2. FIC17003 CASCADE SETPOINT IS SELECTED VIA HS17003A TO AN OPERATOR ENTERED RATIO SETPOINT OR THICKENER CONTROL CUBE.
3. FLOCCULANT DELIVERY TRUCK COMES WITH POWDER BLOWER AND DN100 MALE CAMLOCK ADAPTOR.
4. SIGNALS FROM FLOCCULANT UNLOADING LCS TO DCS.
5. SIREN & BEACON COMBINATION WILL ACTIVATE UPON A HIGH LEVEL TO INDICATE FULL FLOCCULANT SILO.
6. TEMPERATURE SWITCH INSTALLED IN PUMP STATOR BY VENDOR.
7. FUTURE SWAP OVER FROM SCHEME WATER TO BORE WATER FOR FLOCCULANT PLANT.

SET AT 600 kPag
PRV 72051
72-2189-FL-MS1-80
TO GRADE
MS1
PE8
17-0371-FL-PE8-90
FLOTATION TAILINGS THICKENER
2000-F17-PID-PR-10001
FLOCCULANT
80VCD1C1 F72-12169
80VBD1C1 F72-12170
80VBD1C1 F72-12172
25VB03S1 F72-12171
72-1230-FL-MS1-80
TAH 72053
TSH 72053
TE 72053
NOTE 7
DN50
NOTE 3
VSD
15VB03S1 F72-12120
72-1228-FL-MS1-100
100VBD1C1 F72-12164
100VBD1C1 F72-12166
100x80
F72-PP-691
FLOTATION TAILINGS THICKENER FLOCCULANT DOSING PUMP 1
100VBD1C1 F72-12166
25VB03S1 F72-12165
25VB03S1 F72-12167
25VB03S1 F72-12165
SP MC-001
SET AT 600 kPag
PRV 72052
72-2184-FL-MS1-80
80VCD1C1 F72-12178
80VBD1C1 F72-12179
80VBD1C1 F72-12181
25VB03S1 F72-12180
72-1231-FL-MS1-80
TAH 72054
TSH 72054
TE 72054
NOTE 7
DN50
NOTE 3
VSD
15VB03S1 F72-12121
72-1229-FL-MS1-100
100VBD1C1 F72-12173
100VBD1C1 F72-12175
100x80
F72-PP-692
FLOTATION TAILINGS THICKENER FLOCCULANT DOSING PUMP 2
25VB03S1 F72-12174
25VB03S1 F72-12176
SP MC-001
72-4061-FL-MS1-50
CONCENTRATE PRE-LEACH THICKENERS FLOCCULANT DOSING PUMP
2000-F72-PID-PR-10016
FLOCCULANT
72-0972-FL-MS1-100
CONCENTRATE THICKENERS FLOCCULANT DOSING PUMPS SKID
2000-F72-PID-PR-10016
FLOCCULANT
72-1076-FL-PE8-90
FLOTATION TAILINGS THICKENER RECIRCULATION FEED BOX
2000-F17-PID-PR-10001
SPILLAGE

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 – PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 – PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 – PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 – PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 – PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 – PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 – PIPING AND INSTRUMENTATION DIAGRAM

REV.	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	05SEP25	GENERAL REVISIONS AS NOTED	[redacted]				
1	10JUL25	GENERAL REVISIONS AS NOTED					
0	02DEC24	APPROVED FOR CONSTRUCTION					

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PRIMERO PROJECT NUMBER: 23517

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: REAGENTS MIXING & STORAGE
REAGENTS – FLOCCULANT – SHEET 1
PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F72-PID-PR-10004	2

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